

Repository of Experimental Datasets for Spectroscopic Techniques

Pedro Vieira¹, Diogo Cachetas¹, Miguel Rocha¹ e Marcelo Maraschin²

¹ University of Minho, Braga 4710-057, Portugal

² Federal University of Santa Catarina, Florianópolis 88040-900, Brazil
pg59764@uminho.pt, id11647@alunos.uminho.pt, mrocha@di.uminho.pt, m.maraschin@ufsc.br

Código de campo alterado

Código de campo alterado

Abstract. High-throughput metabolomics heavily relies on spectroscopic techniques, such as Infrared (IR), Raman, and Ultraviolet-Visible (UV-Vis) spectroscopy, to comprehensively characterize biological systems. However, despite their widespread application, the management of spectroscopic data significantly lags behind other analytical methods like mass spectrometry. Current data-sharing practices predominantly rely on general-purpose repositories that lack domain-specific metadata validation. This results in fragmented, "metadata-blind" datasets that fail to capture critical experimental contexts (e.g., sample pH, laser wavelength, or sampling mode), thereby hindering cross-study integration and advanced machine learning applications. To address this critical infrastructure gap, this work proposes the development of a centralized, domain-specific repository for spectroscopic datasets. By enforcing a rigorous, modality-specific metadata schema towards harmonized data integration, the proposed framework ensures that all stored spectra adhere to FAIR (Findable, Accessible, Interoperable, and Reusable) principles. Ultimately, this infrastructure aims to standardize spectroscopic data management, providing a highly curated foundation for integrative multi-omics analysis and advanced chemometric pipelines.

Keywords: Repository, IR, Raman, UV-Vis

1 Introduction

While initial research heavily focused on genomics, current paradigms emphasize integrative multi-omics approaches [1]—critically including metabolomics—to fully capture the dynamic complexity of biological systems. Within this framework, metabolomics has emerged as a crucial discipline. By providing comprehensive qualitative and quantitative data—enabling the precise identification and chemical characterization of metabolites—it allows researchers to evaluate specific compounds of interest across diverse biological matrices and organisms [2]. However, the comprehensive characterization of these functional phenotypes relies entirely on advanced analytical techniques, positioning spectroscopy as a fundamental tool for high-throughput metabolic profiling.

Spectroscopy, the study of the interaction of electromagnetic radiation with matter [3-8], has multiple modalities, e.g. Infrared (IR), Raman, Ultraviolet-Visible (UV-Vis).

However, despite their extensive application in scientific literature, the management of spectroscopic data significantly falls behind other analytical methods. Mass spectrometry (MS) benefits from well-established data standards and many robust centralized repositories, such as MetaboLights [9], MassIVE [10], PRIDE [11], NIST Mass Spectrometry Data Center [12], Metabolomics Workbench [13] and MassBank [14] which enforces strict curation of both raw data and experimental metadata. In contrast, the absence of standardized metadata in spectroscopic datasets (IR, Raman and UV-Vis) prevents the identification of molecular patterns across different omics layers. Although analyses limited to a single data type provide valuable insights, literature highlights that integrative multi-omics frameworks are significantly more effective in elucidating causative biological mechanisms [1]. Consequently, the fragmentation of spectroscopic data hinders the development of the integrative models required for a comprehensive understanding of biological systems.

To address these limitations, this work proposes the development of a standardized infrastructure for the management and integration of diverse spectroscopic datasets. The primary objective is to create a robust repository capable of consolidating biochemical and metabolomic data, such as Nuclear Magnetic Resonance (NMR), Gas chromatography–mass spectrometry (GC-MS), and Liquid chromatography–mass spectrometry (LC-MS) with spectroscopic modalities such as IR, Raman, and UV, particularly focused on complex matrices and natural products. This repository will implement data curation standards based on established reference databases like the aforementioned, ensuring the inclusion of critical metadata. Furthermore, the infrastructure will be integrated with the specmine package [15] and the webspecmine application [16], providing a unified pipeline for the storage, retrieval, and analysis of multi-platform experimental data.

2 State of the art

2.1 Spectroscopic Techniques

Infrared (IR)

All organic compounds and many inorganic compounds absorb IR radiation [17]. The scanning of a molecular sample with successive infrared frequencies enables the extraction, qualitatively, of mechanical vibration frequencies of the molecule [18].

IR spectroscopy remains a cornerstone of analytical metabolomics due to several inherent strengths. First, the technique offers a significantly higher signal-to-noise ratio and sensitivity compared to Raman scattering, which is a fundamentally weaker physical process. This efficiency allows for rapid data acquisition, a critical factor in high-throughput screening. Furthermore, IR spectroscopy benefits from extensive spectral libraries (e.g. NIST and Wiley), facilitating the identification of unknown compounds through established reference standards [19]. The adoption of Attenuated Total Reflectance (ATR) sampling has further enhanced the technique's utility, allowing for the direct analysis of diverse biological matrices with minimal preparation, thereby

reducing experimental artifacts and improving reproducibility. Finally, IR is particularly sensitive to changes in the molecular dipole moment, providing superior detection of polar functional groups, such as carbonyls and amides, which are essential for characterizing protein secondary structures and lipid profiles in complex natural products [19-20].

A major bottleneck in Infrared (IR) spectroscopy for biomedical applications is the intense absorption of radiation by water molecules. Due to the high polarity of the O-H bond, water exhibits dominant absorption bands which obscure critical biological signals such as the Amide I band of proteins [19].

Raman

Similarly to IR, Raman spectroscopy analyzes molecular vibrations; however, it is based on the inelastic scattering of monochromatic laser light upon interaction with molecules, resulting in a Raman shift. These frequency shifts correspond to the excitation of specific vibrational levels, providing detailed information regarding molecular structure and conformation [21]. A critical advantage of Raman spectroscopy over infrared techniques is its insensitivity to water interference, making it exceptionally suitable for analyzing biological samples with high water content in their native state [22-23].

Beyond its compatibility with aqueous environments, Raman spectroscopy offers superior spatial resolution, often reaching the sub-micron scale. This allows for "chemical imaging" or Raman microscopy, enabling the mapping of metabolite distribution within single cells or complex tissue structures without the need for extrinsic labels [24]. Furthermore, the use of confocal Raman microscopy allows for non-invasive optical sectioning, providing depth profiles of heterogeneous samples, which are particularly relevant for natural products and multilayered biological matrices [25].

The primary drawbacks of Raman spectroscopy is its inherent weakness, small number of photons undergoes Raman scattering. This often leads to low sensitivity and long acquisition times to achieve an acceptable signal-to-noise ratio [24]. Additionally, biological samples frequently exhibit strong autofluorescence, which can be several orders of magnitude more intense than the Raman signal, effectively hindering the spectral data information [26].

UV-Vis

Unlike vibrational techniques, UV-Vis spectroscopy measures the absorption of light resulting from electronic transitions within molecules, typically involving π -electrons or non-bonding electrons (n-electrons). This technique is fundamental for detecting conjugated systems, aromatic compounds, and certain metal complexes in biological matrices [27].

UV-Vis spectroscopy excels in its robustness for quantification, governed by the Beer-Lambert Law, which establishes a linear relationship between absorbance and

analyte concentration. This makes it an essential tool for determining the concentration of proteins (e.g., at 280 nm) and nucleic acids (e.g., at 260 nm), as well as monitoring enzymatic reactions in real-time [28]. Furthermore, UV-Vis instrumentation is highly accessible, cost-effective, and easily integrated into automated pipelines, such as HPLC-coupled detectors, providing a high degree of reliability for routine laboratory analysis [29].

One of its main limitations of UV-Vis spectroscopy is its low molecular specificity. Because many different compounds can have broad, overlapping absorption bands in the UV-Vis range, it is often difficult to identify specific metabolites in complex mixtures without prior chromatographic separation [27]. Additionally, the technique is highly sensitive to the chemical environment, where factors such as pH, temperature, and solvent polarity can cause shifts in the absorption maximum potentially leading to inaccurate quantifications if these variables are not controlled [29].

2.2 Metadata Frameworks

The scientific utility of spectroscopic datasets is intrinsically linked to the rigor of their associated metadata. As highlighted by integrative multi-omics frameworks, the absence of standardized experimental context—such as sample preparation, instrument settings, and environmental variables—renders raw spectra nearly impossible to replicate or integrate into larger cross-study analyses [1, 19].

2.2.1 Technique-Specific Requirements

Each spectroscopic modality demands a tailored metadata schema to account for their different necessary criteria. In the case of Raman, due to the high sensitivity of the Raman effect to the chemical environment, the documentation of pH levels is critical, as protonation states can significantly shift vibrational bands. Furthermore, the laser excitation wavelength and incident power must be recorded to account for varying fluorescence backgrounds and potential thermal degradation of biological samples [26].

Infrared (IR) Spectroscopy: Metadata must specify the sampling mode (e.g., ATR vs. transmission) and the physical state of the sample (e.g., hydrated tissue vs. lyophilized powder), as these factors dictate the effective path length and the degree of water interference [20].

UV-Vis Spectroscopy: Following the Beer-Lambert law, the optical path length and solvent polarity are mandatory fields. Without these, quantitative comparisons of molar absorptivity across different experimental setups are mathematically invalid [29].

2.2.2 Landscape of Existing Repositories and Data Acquisition

Despite the increasing volume of open-access spectral data, most general-purpose repositories, such as Zenodo [30] or Figshare [31], lack domain-specific metadata validation. In these platforms, most data is scattered in different formats, showcasing the lack of standard (e.g., proprietary vendor files or non-standardized CSVs). On the other

hand, specialized multi-omics repositories like MetaboLights [9] focus predominantly on MS leaving a significant infrastructure gap for spectroscopy techniques. The proposed repository allows to fill this void by enforcing a metadata-first approach, ensuring that every stored spectrum is "FAIR" (Findable, Accessible, Interoperable, and Re-usable) for both humans and machine learning algorithms.

Analyzing recent deposits on Zenodo reveals a wide variety of data, ranging from forensic analysis of hair dye colorants via Raman spectroscopy [32], near-infrared Raman analysis of redyed fabrics [33], materials science matrices such as battery inter-phase components via Attenuated total reflectance-Fourier transform infrared (FTIR) spectroscopy (ATR-FTIR) [34], to UV-Vis analyses of botanical extracts [35]. The absence of a centralized repository that categorizes these biological datasets according to their acquisition methodologies underscores the critical need for the domain-specific infrastructure proposed in this project

To address this challenge of data fragmentation and formatting, recent computational frameworks have begun to emerge. For instance, the Python-based library RamanSPy includes a dedicated datasets module that curates and standardizes programmatic access to a selection of valuable open-source Raman datasets, facilitating their direct use in machine learning workflows [36].

2.3 Integration with the specmine ecosystem

In the broader landscape of spectroscopic analysis, several robust computational tools have emerged within the R ecosystem. Notable examples include hyperSpec [37], which is highly optimized for managing hyperspectral imaging and Raman datasets, and ChemoSpec [38], which provides extensive tools for exploratory chemometrics (e.g., Principal Component Analysis (PCA) and Hierarchical Cluster Analysis (HCA)) across various spectroscopic modalities. Although, these packages typically operate as isolated local environments that require significant manual data wrangling.

To improve and unify the analysis of such data, in collaboration with UFSC in Brazil, the CEB's BiSBi research group has been developing the specmine framework [15]. This comprehensive, open-source R package and web-based application (web-specmine) is designed for the chemometric and machine learning analysis of both multi-omics and spectroscopic data. It equips researchers with a robust suite of analytical tools, which are broadly categorized into three domains: automated spectral preprocessing (e.g., baseline correction, smoothing, and peak alignment), exploratory data analysis (such as Principal Component Analysis and clustering), and predictive modeling (e.g. Partial Least Squares Discriminant Analysis and Random Forests). These diverse methods have been utilized to create various analytical pipelines for metabolomics (GC-MS and NMR) and spectroscopic (IR, UV-Vis) data. However, despite its advanced analytical pipeline, a significant constraint remains in the data ingestion phase. Currently, the specmine package demands manual work, relying heavily on the importation of local CSV files.

Data Harmonization

To rearrange and organize the data, a dedicated Extract, Transform, and Load (ETL) pipeline will be conceptualized. The primary objective of this ETL process is data harmonization—converting diverse, unstructured formats and unstandardized metadata into a uniform, FAIR-compliant structure before ingestion into the repository.

A core architectural principle driving this development is the adoption of a minimal dependency framework. In the context of bioinformatics and software engineering, heavy reliance on third-party libraries often introduces structural vulnerabilities. In such scenarios, future updates, deprecations, or conflicts within external packages can inadvertently break the data pipeline. To mitigate these risks, the Extract, Transform, and Load (ETL) processes—particularly the metadata parsers and harmonization scripts—will be developed primarily using native, custom-built functions. By minimizing the reliance on external dependencies, this architectural choice ensures the long-term stability, reproducibility, and maintainability of the repository, guaranteeing its resilience as a robust backend for the *specmine* ecosystem.

3 References

1. Hasin Y, Seldin M, Lusis A (2017) Multi-omics approaches to disease. *Genome Biol* 18: 83.doi:10.1186/s13059-017-1215-1
2. Zaikin, V.G., Borisov, R.S. Mass Spectrometry as a Crucial Analytical Basis for Omics Sciences. *J Anal Chem* 76, 1567–1587 (2021). doi:10.1134/S1061934821140094
3. Pérez-Juste I, Nieto Faza O (2015) Interaction of radiation with matter. In: Cid MM, Bravo (eds) *Structure elucidation in organic chemistry: the search for the right tools*. Wiley-VCH, Germany, pp 1-25
4. Struve WS (1989) *Fundamentals of molecular spectroscopy*. Wiley-Interscience, New York
5. Herzberg G (1950) *Molecular spectra and molecular structure, I. Spectra of diatomic molecules*. D. Van Nostrand, New York
6. Levine IN (1975) *Molecular spectroscopy*. John Wiley, New York
7. Banwell CN (1994) *Fundamentals of molecular spectroscopy*, 4th edn. McGraw-Hill, London
8. Walker S, Straughan BP (1976) *Spectroscopy*. Chapman and Hall, New York
9. MetaboLights (2024) Repository for metabolomics data. <https://www.ebi.ac.uk/metabolights/>. Accessed 9 Apr 2026
10. MassIVE (2024) Massive spectrometry interactive virtual environment. <https://massive.ucsd.edu/>. Accessed 9 Apr 2026
11. PRIDE Archive (2024) Proteomics identifications database. <https://www.ebi.ac.uk/pride/archive/>. Accessed 9 Apr 2026

12. NIST (2024) NIST mass spectrometry data center. <https://chemdata.nist.gov/>. Accessed 9 Apr 2026
13. Metabolomics Workbench (2024) National metabolomics data repository. <https://www.metabolomicsworkbench.org/>. Accessed 9 Apr 2026
14. MassBank (2024) High resolution mass spectral database. <https://mass-bank.eu/MassBank/>. Accessed 9 Apr 2026
15. Costa RS, Rocha M (2016) Specmine: A framework for molecular spectroscopy data processing and machine learning. *Chemometrics and Intelligent Laboratory Systems* 149:1–9. doi: 10.1016/j.chemolab.2015.10.003
16. Cardoso S, Costa RS, Rocha M (2019) WebSpecmine: a web site for the analysis of metabolomics and spectroscopy data. *Bioinformatics* 35(21):4510–4512. doi:10.1093/bioinformatics/btz285
17. Gupta P, Das SS, Singh NB (2023) *Spectroscopy*. Narosa Publishing House / Taylor & Francis. <https://doi.org/10.1201/9781003412588>
18. Parker F (1971) *Applications of infrared spectroscopy in biochemistry, biology, and medicine*. Springer US. <https://doi.org/10.1007/978-1-4684-1872-9>
19. Griffiths PR, de Haseth JA (2007) *Fourier transform infrared spectrometry*, 2nd edn. Wiley-Interscience, Hoboken.
20. Baker MJ, Trevisan J, Bassan P, et al (2014) Using Fourier transform IR spectroscopy to analyze biological materials. *Nature Protocols* 9(8):1771–1791. doi:10.1038/nprot.2014.110
21. Aroca R (2006) *Surface-enhanced vibrational spectroscopy*. Wiley, Chichester
22. Cachetas DM (2023) Synergizing artificial intelligence with water monitoring. Master's Thesis, University of Minho, Braga
23. Bi X, et al (2025) Artificial intelligence-powered surface-enhanced Raman spectroscopy for biomedical applications. *Analytical Chemistry* 97(13):6826–6846. <https://doi.org/10.1021/acs.analchem.4c06584>
24. Eberhardt K, Stiebing C, Matthäus C, et al (2015) Advantages and limitations of Raman spectroscopy for molecular diagnostics. *Expert Review of Molecular Diagnostics* 15(6):773–787. doi:10.1586/14737159.2015.1036776
25. Butler HJ, Ashton L, Bird B, et al (2016) Using Raman spectroscopy to characterize biological materials. *Nature Protocols* 11(4):664–687. doi:10.1038/nprot.2016.036
26. Schie IW, Huser T (2013) Methods and applications of Raman spectroscopy. *Applied Spectroscopy* 67(8):813–828. doi:10.1366/12-06971
27. Schmid FX (2001) Biological macromolecules: UV-visible spectrophotometry. *Encyclopedia of Life Sciences*. doi:10.1038/npg.els.0003142
28. Gore MG (2000) *Spectrophotometry and spectrofluorimetry: a practical approach*. Oxford University Press, Oxford
29. Pavia DL, Lampman GM, Kriz GS, Vyvyan JR (2014) *Introduction to spectroscopy*, 5th edn. Cengage Learning, Stamford
30. European Organization for Nuclear Research, OpenAIRE (2013) Zenodo. <https://zenodo.org>. Accessed 9 Apr 2026
31. Figshare (2011) Figshare repository. <https://figshare.com>. Accessed 9 Apr 2026

32. Holman AP, Maalouf A, Kurouski D (2026) Dataset for: DyeSPY: Establishing the First Forensic SERS Reference for Hair Dye Colorant Evidence. Zenodo. <https://doi.org/10.5281/zenodo.18881839>
33. Bober S, Kurouski D (2026) Dataset for: Near-infrared excitation Raman analysis of Underlying colorants on redyed fabric. Zenodo. <https://doi.org/10.5281/zenodo.18881215>
34. Karapın-Springorum L, et al (2026) Database of ATR-FTIR, Raman spectroscopy, and X-ray diffraction reference data for battery interphase components. Zenodo. <https://doi.org/10.5281/zenodo.18553569>
35. Slavov A, et al (2020) UV-Vis spectral dataset of distillation wastewaters from the production of essential oils of lavender cultivars and other aromatic plant species. Zenodo. <https://doi.org/10.5281/zenodo.4407442>
36. Georgiev D, et al (2024) RamanSPy: Datasets. RamanSPy Documentation. <https://ramanspy.readthedocs.io/en/latest/datasets.html>. Accessed 9 Apr 2026
37. Beleites C, Sergo V (2021) hyperSpec: a package to handle hyperspectral data sets in R. R package version 0.100.0. <https://r-hyperspec.github.io/hyperSpec/>
38. Hanson BA (2022) ChemoSpec: Exploratory Chemometrics for Spectroscopy. R package version 6.1.9. <https://bryanhanson.github.io/ChemoSpec/>